

SAMPLE REPORT



Autonomous Mobile Robot (AMR)

MARKET ANALYSIS, 2017 – 2030

Opportunities Beyond COVID-19 Crisis

BASE YEAR: 2021

TECHNOLOGY

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Autonomous Mobile Robot (AMR)

MARKET ANALYSIS, 2017 - 2030

Market by Component: Hardware, Software, Services

Market by Type: Goods-to-Person Picking Robots, Self-Driving Forklifts, Autonomous Inventory Robots, Unmanned Aerial Vehicles

Market by Battery Type: Lead Battery, Lithium-Ion Battery, Nickel-based Battery, Others

Market by End Use: Manufacturing, Wholesale and Distribution

Market by Manufacturing: Automotive, Aerospace, Electronics, Chemical, Pharmaceuticals, Plastics, Defense, FMCG, Others

Market by Wholesale & Distribution: E-commerce, Retail Chains/ Conveyance Stores, Others

Market by Region: North America, Europe, Asia Pacific, South America, Middle East & Africa

TECHNOLOGY INDUSTRY EXPERTS:

Principal Author



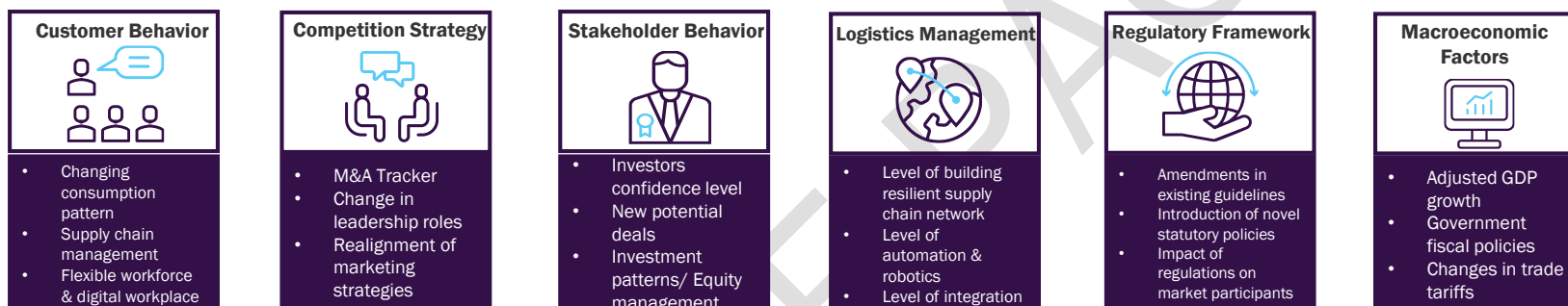
Jagdish Thadani

Jagdish oversees the operation of the technology team, from resource training & development (T&D) and project management to devising new research methodologies on a case-to-case basis. He is also actively engaged with clients and strives towards building open, productive relationships. He leads a team of market research analysts to support tier-1 technology vendors with Grand View Research's market research and advisory services. His key role involves market sizing, forecasting, and delivering insights to key clients.

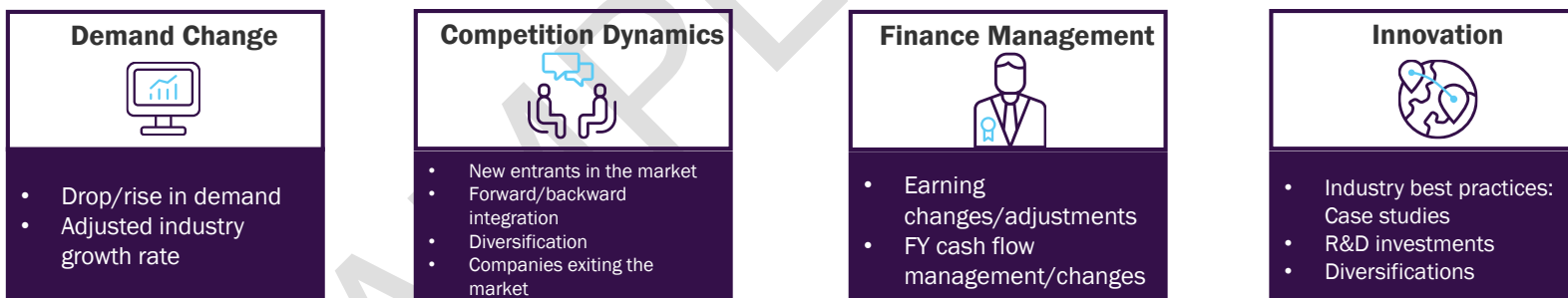
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How is the impact of COVID-19 Covered in the Report

Response Based Factors Covered in the Report



Outcome Based Factors Covered in the Report





Market Voice : Key Opinions

Analyst Perspective

“AMRs are highly adopted for warehouse operations due to its numerous benefits, including reduced product damage, reduced labor costs, upscaled productivity, and automation processes. These notable advantages of autonomous mobile robots are being witnessed across several industries, including transportation & logistics, automotive, and food & beverages, among others.”

CEO of AutoGuide Mobile Robots: A Teradyne Company

“Looking at AMR vs. conveyor networks or some traditional line guided vehicles, it offers flexibility, scalability, safety, and much faster to deploy. Also, when moving material around in the past, you had fixed conveyor systems or even fixed-line guided vehicle systems. Today with AMRs, you can actually expand your system throughout your factory without having to do any infrastructure changes.”

Note: Above listed verbatim responses are intended to provide an overview of the market pulse. Please get in touch with us to learn more about our primary research methodology & how we could customize our services to meet your end objective.



Top 3 Trends : Key Takeaways

- 1 { The AMR market is witnessing high competition among robot manufacturers as they are taking various strategic initiatives to gain a competitive edge in the robotics industry. AMRs reduce person-to-person interaction in warehouses, making them a safer alternative for your staff and their loved ones in light of the latest pandemic.
- 2 { Transportation & logistics industry are particularly deploying AMRs to boost production and distribution operations in line with the increasing demand. With the fast-developing e-commerce business leading to direct delivery trends, order fulfillment productivity has become a priority for warehouse operators
- 3 { Autonomous Mobile Robots (AMRs) have triggered a paradigm shift in performing tasks that are commonly associated with manufacturing and distribution, such as material picking, handling, and sorting

Sample Navigator

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What does it include?

This section will help you access the structure of the report & research scope in detail

Section wise description provided for clarity

How will it benefit me?

You will be able to judge the final product deliverables and map them against your requirements

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What does it include?

This section includes sample market data points, ranging from trend analysis to market estimates & forecasts

How will it benefit me?

You will gain access to a synopsis of the market report.

Sample data points will help you validate market estimates & judge report quality

Research Methodology

What does it include?

A detailed account of our research methodology

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Report FAQs

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Executive Summary

Market Insights

The adoption of material handling automation solutions for inventory management is growing significantly across various industries.

E-commerce companies deploy autonomous mobile robots in their inventories and proliferate their warehouse operations by performing acquisitions of large-scale autonomous robots' vendors.

The Asia Pacific region is anticipated to witness significant growth in the AMR market.

Autonomous mobile robots are used to pick, transport, and sort items in manufacturing and distribution facilities without any manual intervention. AMRs can perform various operations, such as moving raw materials and manufactured goods within the facility, by utilizing a bundle of vision cameras, onboard sensors, and facility maps integrated with the warehouse execution software (WES). AMRs are highly adopted for warehouse operations due to its numerous benefits, including reduced product damage, reduced labor costs, upscaled productivity, and automation processes. These notable advantages of autonomous mobile robots are being witnessed across several industries, including transportation & logistics, automotive, and food & beverages, among others.

In the warehousing sector, **retailing companies** are collaborating with tech companies and automation solutions providers to transform material handling operations. For instance, in June 2021, DHL Supply Chain and Locus Robotics came to an agreement to add 2,000 AMRs to DHL by the end of 2022. As of September 2022, Locus Robotics announced that the company's current investor, Tiger Global Management, committed USD 50 million. Formerly in February 2021, Tiger Global Management and Bond, an IT investment firm, had partnered to raise USD 150 million in a Series E round.

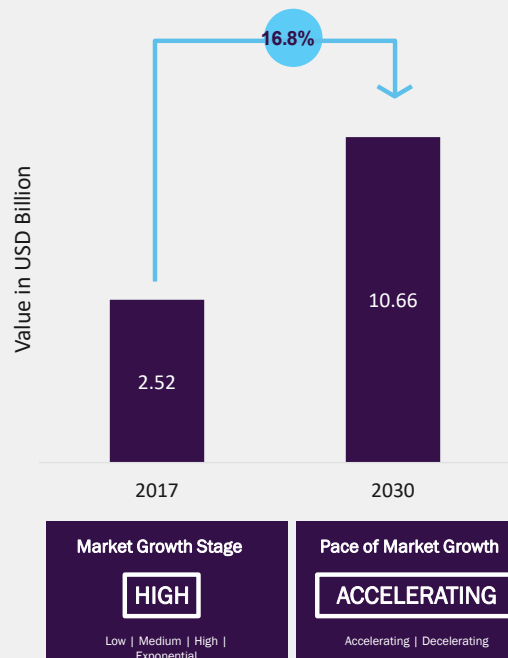
The **lead battery segment** accounted for the largest market share in 2021 due to the low cost of these batteries. Lead batteries also ensure stable voltage, good reversibility, and long service life and are hence well-suited for a broad range of industrial applications. For instance, Kuka AG's KMR QUANTEC AMR mobile industrial robot system can be considered as a prominent example of lead battery-based AMRs. It leverages the industrial wireless LAN technology, thereby eliminating the need for cabling while offering a higher degree of adaptability.

MARKET SNAPSHOT

Fig.1 Market snapshot, 2021

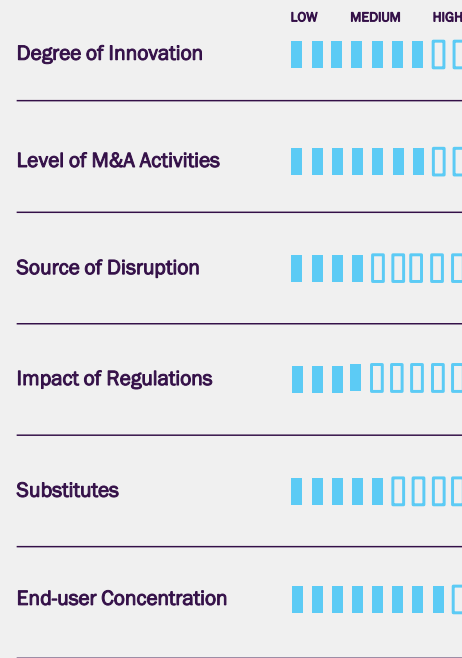
Market Facts

Market Size



Market Characteristics

Market Attributes



Key Takeaways

Market Driver

- Emergence of artificial intelligence technology in autonomous mobile robots
- High adoption of robots in manufacturing and warehouse logistics operations

Market Restraint

- Variation in application-specific requirements

Key Vendors

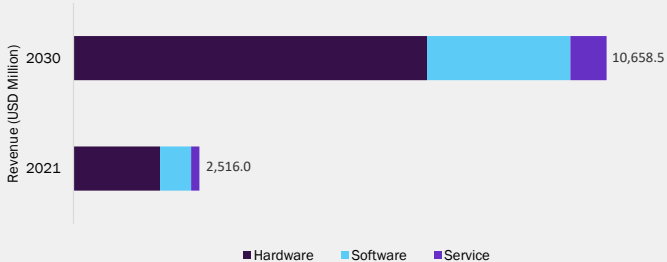
- ABB
- Bleum
- Boston Dynamics
- Clearpath Robotics, Inc.
- GreyOrange
- Harvest Automation
- IAM Robotics
- inVia Robotics, Inc.
- KUKA AG
- Teradyne Inc.

Source: BJET, ETS, Education Technology Magazine, Design and Technology Education: An International Journal, IEEE, Company Annual Reports, Grand View Research

Segmental Insights

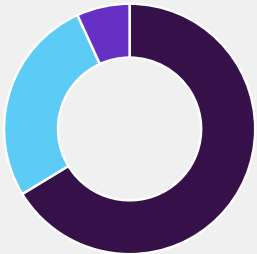
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Market opportunity by Component

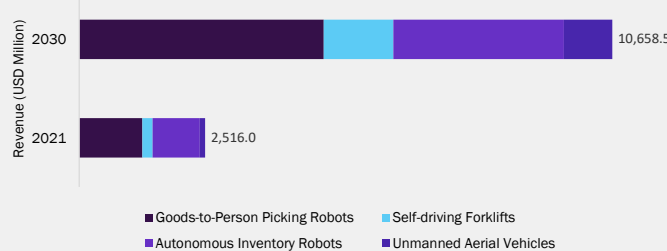


	CAGR% (2022-2030)
Hardware	-
Software	-
Services	-

Market Opportunity*

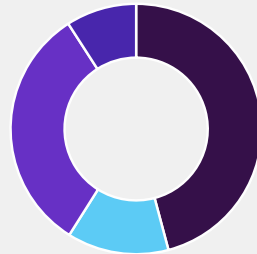


Type Segmentation Analysis
Market opportunity by Type



	CAGR% (2022-2030)
Goods-to-person Picking Robots	-
Self-driving Forklifts	-
Autonomous Inventory Robots	-
Unmanned Aerial Vehicles	-

Market Opportunity*



*Market opportunity represents the market share value for the year 2030

Source: BJET, ETS, Education Technology Magazine, Design and Technology Education: An International Journal, IEEE, Company Annual Reports, Grand View Research

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Section Description:

This section intends to provide an overview of market lineage for the study product market. We believe covering aspects of parent market enables us to provide holistic view on the various connected factors impacting the study market.

This section also entails in-depth analysis on variables related to value chain, technology, regulatory frameworks and other allied market dynamics impacting the study market.

Market Variables, Trends & Scope

Market Lineage Outlook

RELATED MARKET OUTLOOK

AUTOMATED GUIDED VEHICLE MARKET

AGVs are being used in various end use industries, such as logistics, automotive, healthcare, manufacturing, and food and beverages. Increased productivity and reduced labor costs are the key factors driving the need for industrial equipment. The demand for AGVs is expected to grow owing to increasing automation in industries. Increasing industrialization has led to the continuous need for customized material handling and transport solutions for fueled efficiency.

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WAREHOUSE ROBOTICS MARKET

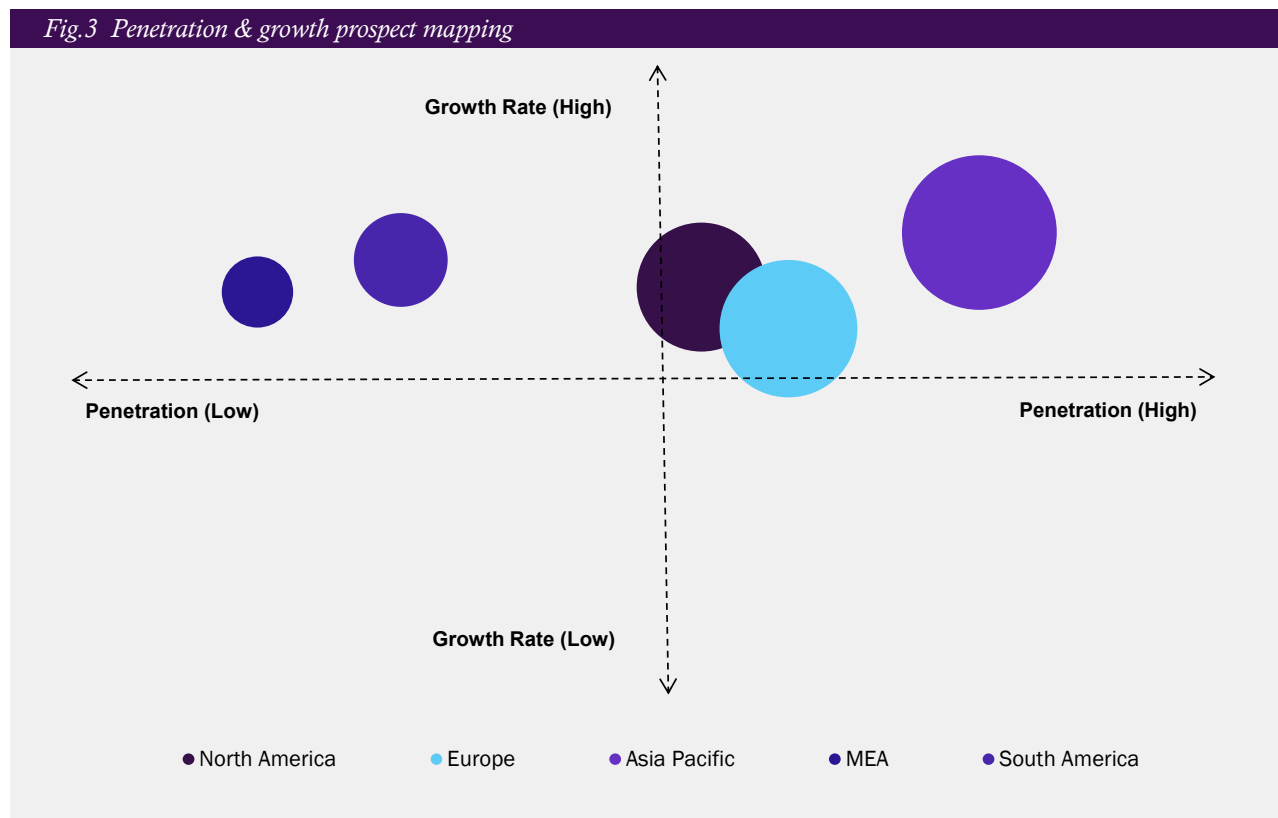
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Market Definition

Autonomous mobile robots are used to pick, transport, and sort items in manufacturing and distribution facilities without any manual intervention. AMRs can perform various operations, such as moving raw materials and manufactured goods within the facility, utilizing a bundle of vision cameras, onboard sensors, and facility maps integrated with the warehouse execution software (WES). AMRs integrated with WES operate with increased flexibility to build their routes between locus within the warehouse and distribution center facility environment. It results in a mobile robot capable of working with facility staff within the dynamic environment.

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Penetration and Growth Prospect Mapping



Source: BJET, ETS, Education Technology Magazine, Design and Technology Education: An International Journal, IEEE, Company Annual Reports, Grand View Research

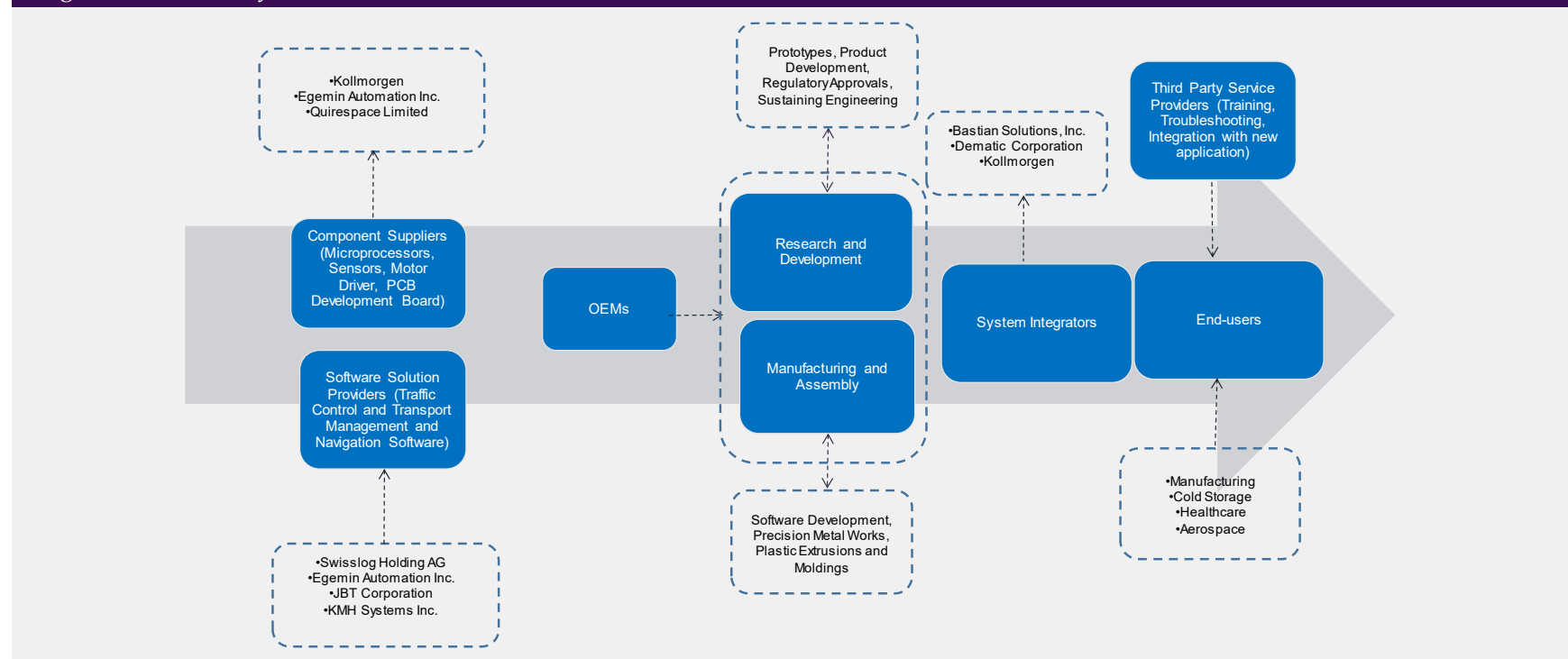
The AMR market is witnessing high competition among robot manufacturers as they are taking various strategic initiatives to gain a competitive edge in the robotics industry. For instance, in September 2021, Locus Robotics, a robotic process automation firm, announced the acquisition of Waypoint Robotics, Inc., a manufacturer of automation machinery. The acquisition aims to expand Locus' product portfolio of tried-and-true AMR solutions, which handle use cases ranging from casepicking, ecommerce, and pallet-picking to scenarios needing heavier, larger payloads and fulfillment modalities.

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Value Chain Analysis

The autonomous mobile robots value chain comprises various ecosystem players, such as component suppliers, software solution providers, system integrators and OEMs, end use vertical and third-party providers. As such, it is highly diversified and fragmented market, with large number of suppliers operating in it. Prominent vendors such as Kollmorgen, Egemin Automation Inc., and Quirespace Limited provide components including microprocessors, sensors, motor driver, and PCB development board. Kollmorgen provides complete hardware and software solutions for adding automated navigation to existing robot designs, as well as incorporates advanced automation capabilities in them.

Fig.4 Value chain analysis



Source: BJET, ETS, Education Technology Magazine, Design and Technology Education: An International Journal, IEEE, Company Annual Reports, Grand View Research

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Section Description:

Every market forecast that we build considers various dimensions of market triggering factors such as product, customer demographics, regulatory, geo-political, technology, economic, financial and competition with varied impacts over stipulated time. These relevant factors & its impacts are closely monitored at each major country level, wherein its sensitivity results in summation to region level forecast and henceforth reflects in global market forecast.

Market dynamics section is a key component of our report evaluating market drivers, restraints and other industry level challenges anticipated to impact growth over forecast period (2030). Below section presents complete list of key factors evaluated during market forecast for your understanding.

Please get in touch with us if you wish to add any additional factors to this list, we will be happy to assist you with your requirement.

Market Dynamics

MARKET DRIVER ANALYSIS

Market Drivers	2017-20	2021-25	2026-30
	Impact		
Emergence of artificial intelligence technology in autonomous mobile robots			
High adoption of robots in manufacturing and warehouse logistics operations			
Increased safety at high-risk workplace			

Key:

Very High	High	Moderate	Low
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Emergence of artificial intelligence technology in autonomous mobile robots

Autonomous mobile robots (AMRs) pick, transport, and sort items within manufacturing and distribution facilities without manual intervention. AMRs use onboard sensors, vision cameras, and facility maps coupled with warehouse execution software (WES) to execute various duties, including moving raw materials and manufactured goods within the facility. AMRs are highly preferred owing to their myriad benefits, such as enhancing productivity, minimizing labor costs, preventing product damage, and automating various processes. AMRs are considered to empower employees by working collaboratively to produce highly productive work environments by automating repetitive and high-risk material transportation. Robots use sensors and algorithms to navigate dynamic environments but cannot take advanced decision-making via sensory inputs. Artificial intelligence is the upcoming addition to the evolution of AMRs to make them intelligent mobile robots. Integrating AI technology into mobile robots makes them more efficient. Many tasks can be performed and are easily adaptable in a working environment. AI enables mobile robots comprised of sensors and software for control and observation.

The robots collect data from 3D cameras, laser scanners, accelerometers, gyroscopes, and wheel encoders to make efficient decisions in critical situations. Robots can move efficiently and use their knowledge of their surroundings to avoid obstacles and persons in their path. AMRs with AI capabilities feature strategically placed cameras that function as extended robot sensors. AMRs receive information about a location before entering it and learn to change their behavior accordingly. This means they research specific jobs and avoid impediments such as high-traffic areas at specified times, such as when supplies are regularly delivered and transferred by truck or when there is a lot of human interaction during breaks or shift changes.

High adoption of robots in manufacturing and warehouse logistics operations

Increased safety at high-risk workplace

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MARKET RESTRAINT/ CHALLENGES ANALYSIS

Market Restraint	2017-20	2021-25	2026-30
	Impact		
Variation in application-specific requirements			

Key:

Very High	High	Moderate	Low
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VARIATION IN APPLICATION-SPECIFIC REQUIREMENTS

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Business Environment Analysis

Industry Analysis – Porter's

Porter's Five Forces model helps identify the key competitive forces that are affecting the market. These forces help in determining the structure of the market and the competition level. The initial capital requirement in the Autonomous mobile robot (AMR) industry is very high, which setup cost, R&D cost, marketing, and supply chain cost, and maintenance cost. The new entrant companies require considerable investments to start a company and sustain in the competitive market.

THREAT OF NEW ENTRANTS

BARGAINING POWER OF BUYERS

BARGAINING POWER OF SUPPLIER

THREAT OF SUBSTITUTES

COMPETITIVE RIVALRY

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PEST ANALYSIS

PEST analysis stands for the political, economic, social, and technological landscape of the market. It encompasses the insights on external factors that influence the business's operation to become more competitive in the market.

POLITICAL LANDSCAPE

ENVIRONMENTAL LANDSCAPE

SOCIAL LANDSCAPE

TECHNOLOGY LANDSCAPE

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Major Deals & Strategic Alliances Analysis

Mergers & Acquisitions and Technology Collaborations

Table 1 Autonomous Mobile robots - M&A, collaborations & partnership scenario

Year	Company Name	Business Description
2022	Six34AI	Six34AI, the Israeli company behind MAESTRO, an AI-enabled control tower for indoor mobility management, has signed an agreement to install 200 autonomous mobile robots (AMRs) throughout Musashi Seimitsu's 35 manufacturing plants worldwide. The MAESTRO-powered AMRs from 634AI will automate Musashi Seimitsu's intralogistics operation and its production floors.
2022	BlueBotics	BlueBotics, a provider of autonomous navigation technology based in Switzerland, was acquired by ZAPI Group, an Italian maker of electronic components. Bluebotics will become a part of a global industrial platform due to this transaction. At the same time, Zapi will be able to advance up the value chain into navigation and logistics services.

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Section Description:

This section intends to provide an overview of market estimates & trends by component offered by vendors in various forms. We have used bottom-up method to arrive at base estimations wherein magnitude of product penetration & revenue generated by the vendors across key countries were taken into consideration. These parameters are integrally built into our forecasting methodology driven by weights assigned to each parameter across short, medium & long term of the forecast.

Inclusion: Component included in the study are hardware, software and services

Exclusion: Not applicable

Autonomous Mobile Robot (AMR) Market: Component Estimates & Trend Analysis

Definition & Scope

We have segmented the global autonomous mobile robot (AMR) market based on following component:

- **Hardware:** With reference to the components, the hardware segment covers the parts used to build the AMRs. These parts comprise controllers, wheels, encoders, brakes, motors, batteries, actuators, sensors, and power systems.
- **Software:** With reference to the components, the software segment covers the programs used to build the AMRs. Autonomous Mobile Robots (AMRs) software includes software components such as scanning and visual processing software, motion planning, navigation systems, control systems, safety systems, fleet management systems, and payload, among other programs.
- **Services:** With reference to the components, the services segment covers the professional and managed services provided by OEMs and third-party service providers. These services may be predictive maintenance and corrective maintenance.

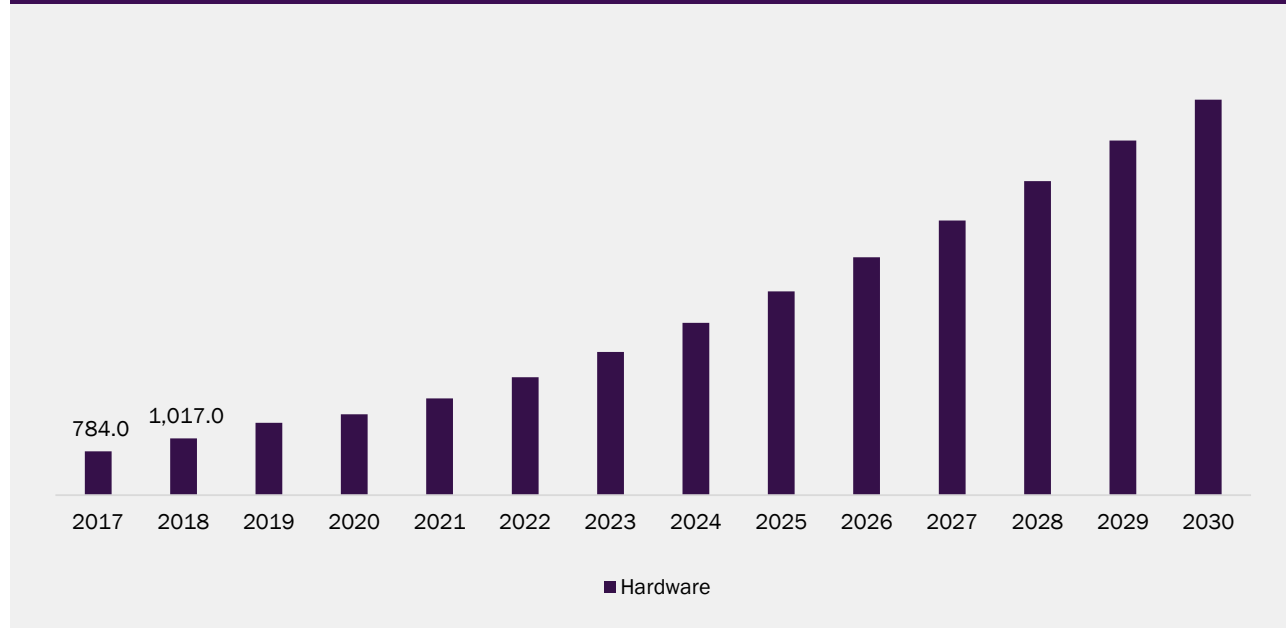
Component movement & market share analysis, 2021 & 2030

Based on component, the autonomous mobile robot market is segmented into hardware, software and services. The hardware segment acquired the dominant revenue share of over 68% in 2021 due to the latest hardware platforms that provide convenient component interconnection and advanced capabilities.

Hardware

Global hardware autonomous mobile robots market estimates and forecasts, 2017–2030, (USD Million)

Fig.5 Hardware autonomous mobile robots market estimates and forecasts, 2017–2030 (USD Million)



Source: AIIM, IEEE, Wired Magazine, IJCT, Technology Blogs, Company Annual Reports, Dun & Bradstreet Hoovers, Grand View Research

The final report will include the aforementioned figure in tabular format.

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Software

Services

Upgrade report license to gain access to the complete analysis of the segment and sub-segment

Section Description:

This section intends to provide an overview of market estimates & trends by type offered by vendors in various forms. We have used bottom-up method to arrive at base estimations wherein magnitude of product penetration & revenue generated by the vendors across key countries were taken into consideration. These parameters are integrally built into our forecasting methodology driven by weights assigned to each parameter across short, medium & long term of the forecast.

Inclusion: Type included in the study are goods-to-person picking robots, self-driving forklifts, autonomous inventory robots, and unmanned aerial vehicles.

Exclusion: Not Applicable

Autonomous Mobile Robot (AMR) Market: Type Estimates & Trend Analysis

Definition & Scope

We have segmented the global autonomous mobile robot (AMR) market based on following type:

- **Goods-to-Person Picking Robots:** These robots are mobile robotic picking solutions that carry carts and navigate flexible routes in the inventory to transport items between workers and stations.
- **Self-Driving Forklifts:** These robots are autonomous trucks used for material handling in an inventory.
- **Autonomous Inventory Robots:** These robots offer inventory monitoring opportunities by combining with RFID-tagged products and equipment in the warehouse.
- **Unmanned Aerial Vehicles:** These robots are used to move large products through the air in distribution centers.

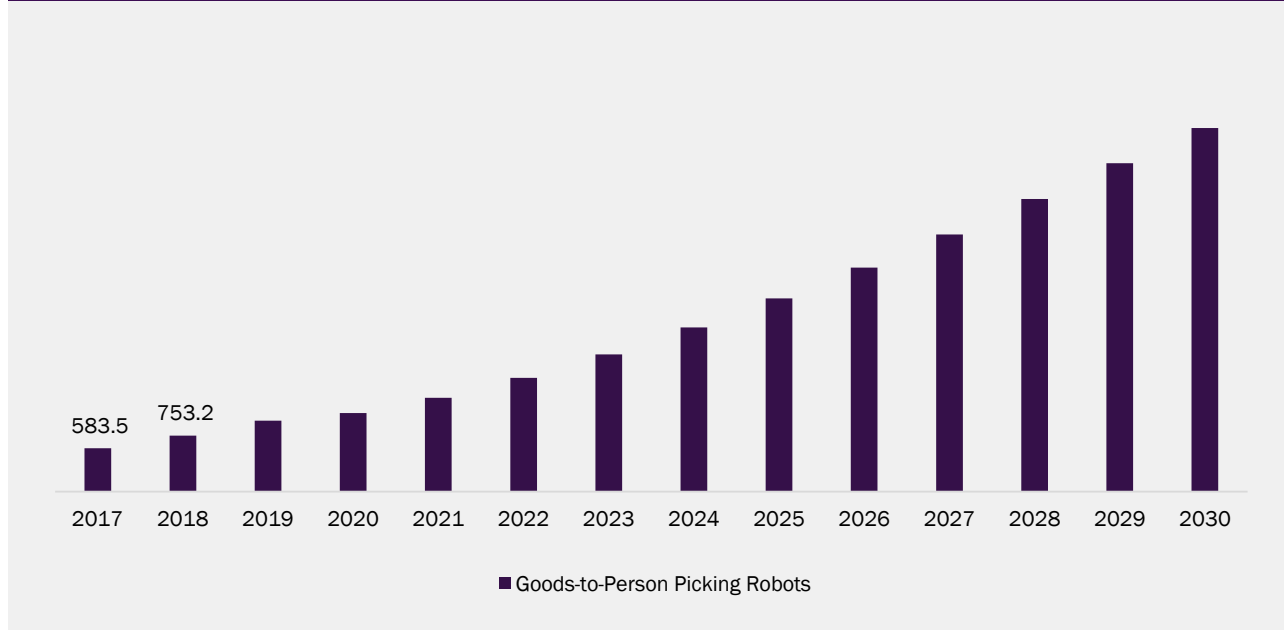
Type movement & market share analysis, 2021 & 2030

The goods-to-person picking robots segment led the market's share in 2021 due to the to the continued deployment of autonomous robots to eliminate paper-based and manual picking systems. Goods-to-person picking robots can be programmed to carry carts and navigate flexible routes to move goods between workers and stations. Several robotics companies, including Bleum, GreyOrange, and IAM Robotics, offer autonomous mobile robotic picking systems that can potentially add a new efficiency level to manufacturing and warehousing operations.

Goods-to-Person Picking Robots

Global goods-to-person picking robots market estimates and forecasts, 2017–2030, (USD Million)

Fig.6 Goods-to-person picking robots market estimates and forecasts, 2017 – 2030 (USD Million)



Source: AIM, IEEE, Wired Magazine, UCET, Technology Blogs, Company Annual Reports, Dun & Bradstreet Hoovers, Grand View Research

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Self-Driving Forklifts

Autonomous Inventory Robots

Unmanned Aerial Vehicles

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Section Description:

This section intends to provide an overview of market estimates & trends by battery type of autonomous mobile robots (AMR). We have used delphi method to arrive at base estimations wherein prevailing trends of AMR based on battery type is considered. These parameters are integrally built into our forecasting methodology driven by weights assigned to each parameter across short, medium & long term of the forecast.

Inclusion: Battery type included in the study are lead battery, lithium-ion battery, nickel-based battery, and others

Exclusion: Not applicable

Autonomous Mobile Robot (AMR) Market: Battery Type Estimates & Trend Analysis

Scope

We have segmented the global autonomous mobile robot (AMR) market based on battery type as following:

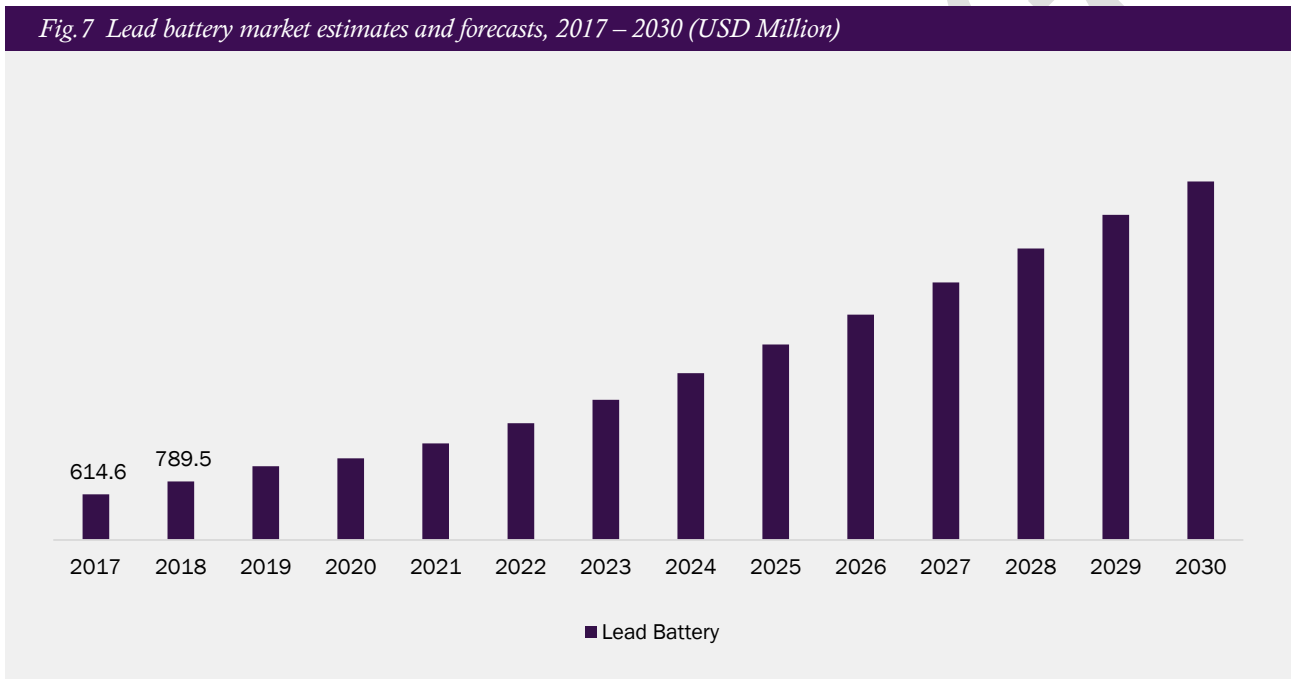
- **Lead Battery:** Lead batteries provide constant voltage, long service life, and good reversibility, making them ideal for various industrial applications. FOR EXAMPLE, Kuka AG's KMR QUANTEC AMR mobile industrial robot system is a renowned example of a lead battery-based AMR.
- **Lithium-Ion Battery:** Lithium-ion batteries are more expensive. However, lithium-ion batteries improve high-power charge and discharge while meeting the needs of many industrial situations, such as high energy density, excellent temperature performance, and a long-life cycle. These benefits are expected to encourage AMR makers to use lithium-ion batteries.
- **Nickel-based Battery:** These batteries are rechargeable, and the voltage rating of each cell is 1.2 V, which comes with a range from 600 mAh to 3300 mAh. The batteries are costly but provide high current capacity and a long-life span.
- **Others:** The others battery type segment includes alkaline batteries and lithium polymer batteries.

Battery type movement & market share analysis, 2021 & 2030

Based on the application, the market has been segmented into lead battery, lithium-ion battery, nickel-based battery and others. The lead battery segment accounted for the largest share in 2021. The high share can be attributed to lead batteries' low-cost advantages over other battery types. Lead batteries also ensure stable voltage, good reversibility, and long service life and are well-suited for a broad range of industrial applications.

Lead Battery

GLOBAL LEAD BATTERY MARKET ESTIMATES AND FORECASTS, 2017–2030, (USD MILLION)



Source: AIIM, IEEE, Wired Magazine, IJCET, Technology Blogs, Company Annual Reports, Dun & Bradstreet Hoovers, Grand View Research

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Lithium-Ion Battery

Nickel-based Battery

Others

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Section Description:

This section intends to provide an overview of market estimates & trends by end use of autonomous mobile robot (AMR). We have used delphi method to arrive at base estimations wherein prevailing trends of AMR based on end use is considered. These parameters are integrally built into our forecasting methodology driven by weights assigned to each parameter across short, medium & long term of the forecast.

Inclusion: End use included in the study are manufacturing and wholesale & distribution

Exclusion: Not applicable

Autonomous Mobile Robot (AMR) Market: End use Estimates & Trend Analysis

Definition & Scope

We have segmented the global autonomous mobile robot (AMR) market based on end use as following:

- **Manufacturing:** This end use is further bifurcated into automotive, aerospace, electronics, chemical, pharmaceuticals, plastics, defense, FMCG, and others.
- **Wholesale and Distribution Sector:** This end use is further bifurcated into e-commerce, retail chains/conveyance stores, and others.

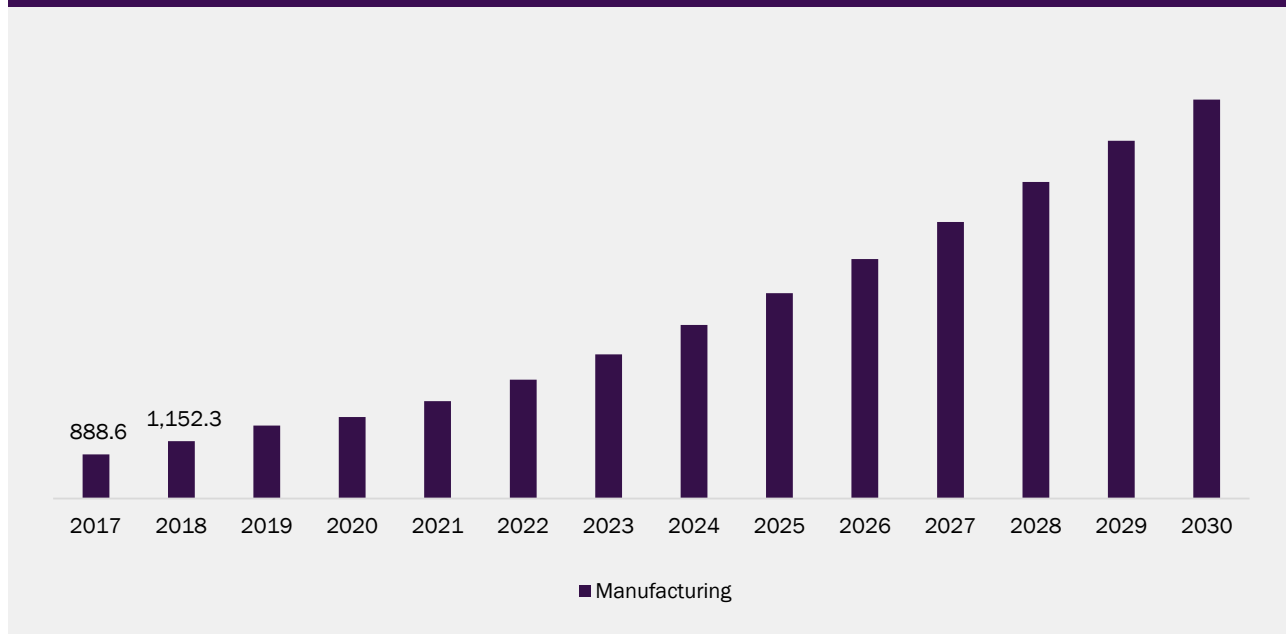
End use movement & market share analysis, 2021 & 2030

The manufacturing segment led the market in 2021 due to the continued process automation in the manufacturing industry worldwide. The segment has been further segmented into automotive, aerospace, electronics, chemical, pharmaceuticals, plastics, defense, FMCG, and others. The automotive segment dominated the market with a high market share owing to the aggressive adoption of AMRs for various tasks, such as assembly, painting, sealing, coating, machine tending & part transfer, and materials removal, as well as internal logistics associated with automotive manufacturing as part of the efforts to introduce lean manufacturing.

Manufacturing

GLOBAL MANUFACTURING MARKET ESTIMATES AND FORECASTS, 2017–2030, (USD MILLION)

Fig.8 Manufacturing market estimates and forecasts, 2017 – 2030 (USD Million)



Source: AIIM, IEEE, Wired Magazine, IJCT, Technology Blogs, Company Annual Reports, Dun & Bradstreet Hoovers, Grand View Research

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Wholesale & Distribution

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Section Description:

This section intends to provide an overview of market estimates & trends by key countries contributing to overall regional revenue estimations for the study product market. We have used bottom-up method to arrive at base estimations wherein product penetration, growth rate, and forecast revenue are captured at key countries level and summed up to respective regions.

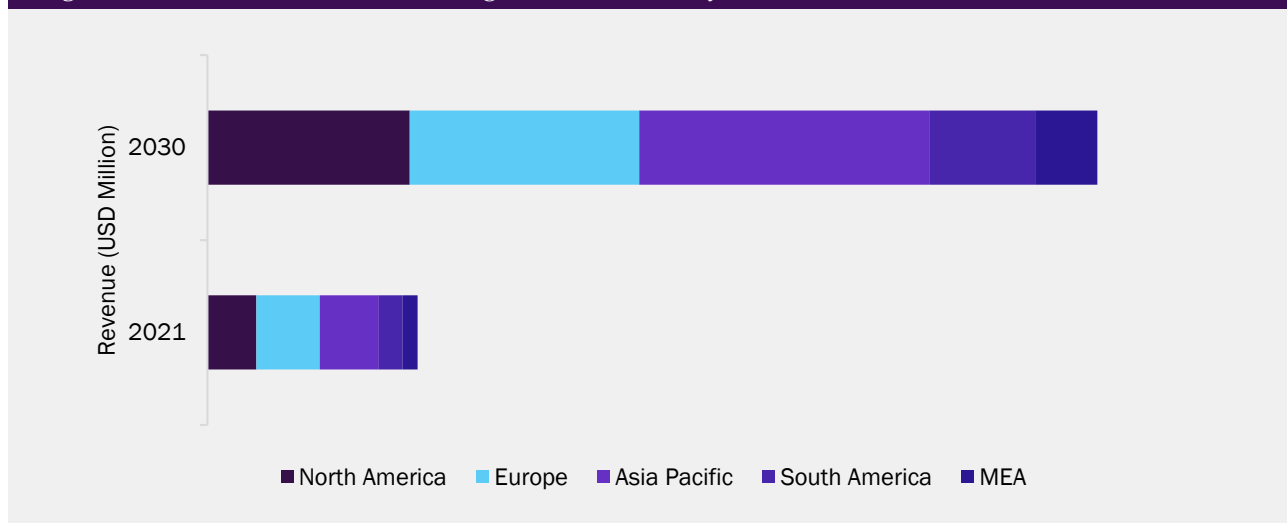
The scope includes data on all major countries spread across North America, Europe, Asia Pacific, South America & Middle East & Africa. Interest in additional countries apart from the major list can be accommodated.

Autonomous Mobile Robot (AMR) Market: Regional Estimates & Trend Analysis

Europe AMR market was valued at USD 759.8 million in 2021 and is expected to witness significant growth over the forecast period, reaching USD 2,749.9 million by 2030, growing at an estimated CAGR of 14.7% over the forecast period.

Regional movement & market share analysis, 2021 & 2030

Fig.9 Autonomous mobile robot market, regional movement analysis, 2021 & 2030



Source: AIIM, IEEE, Wired Magazine, IJCET, Technology Blogs, Company Annual Reports, Dun & Bradstreet Hoovers, Grand View Research

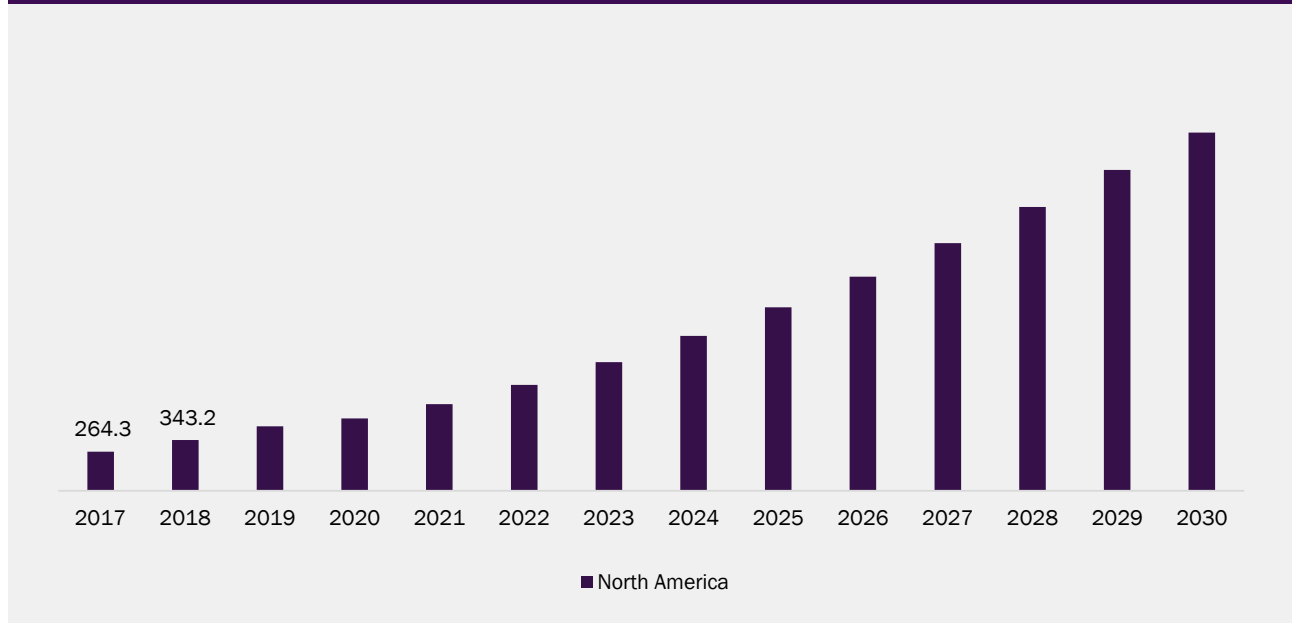
Europe accounted for the largest revenue share in 2021 and is expected to maintain its lead over the forecast period. This high share is attributable to the increasing demand for material handling equipment by the manufacturing industry incumbents. Furthermore, process automation in every industry has also stimulated the market's growth in this region. For instance, in July 2022, 6 River Systems announced establishing a multiyear global deal with GXO Logistics, Inc., a contract logistics provider. Through this partnership, the companies will meet requirements for Collaborative Mobile Robots (CMRs) in their logistics operations throughout the U.S. and Europe.

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North America

North America autonomous mobile robot (AMR) market estimates and forecasts, 2017 – 2030 (USD million)

Fig.10 North America autonomous mobile robot market estimates and forecasts, 2017 – 2030 (USD Million)



Source: AIM, IEEE, Wired Magazine, IJCET, Technology Blogs, Company Annual Reports, Dun & Bradstreet Hoovers, Grand View Research

The final report will include the aforementioned figure in tabular format.

North America autonomous mobile robots market estimates and forecasts, by component, 2017 – 2030 (USD Million)

Table 1 Table North America autonomous mobile robots market, by component, 2017 – 2030 (USD Million)

Type	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	CAGR (2022-30)
Hardware	183.8	-	302.0	-	-	-	-	-	-	-	-	-	-	-	-
Software	-	78.7	-	115.2	-	-	-	-	-	-	-	-	-	-	-
Services	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Note: Similar tables would be provided for all the market segments for all the regions and countries mentioned in the scope

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North America autonomous mobile robot (AMR) market estimates research methodology

Fig.11 North America autonomous mobile robot (AMR) market estimates research methodology



Source: BJET, ETS, Education Technology Magazine, Design and Technology Education: An International Journal, IEEE, Company Annual Reports, Grand View Research

North America

- ▶ **U.S.**
- ▶ **CANADA**
- ▶ **MEXICO**

Europe

- ▶ **U.K.**
- ▶ **GERMANY**
- ▶ **FRANCE**

Asia pacific

- ▶ **CHINA**
- ▶ **INDIA**
- ▶ **JAPAN**

South America

- ▶ **BRAZIL**

Middle East & Africa

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Section Description:

This section intends to provide an overview of competition within market listing major recent developments, company product/ solution offerings and market positioning are discussed in detail. We can also include vendor landscape and distinctive listing for public & private companies active in the market.

Your interest specific to additional data points can be accommodated upon request.

Competitive Landscape

Recent Developments and Impact Analysis, by Key Market Participants

Vendors in the AMR market focus aggressively on expanding their customer base and gaining a competitive edge over their rivals. Hence, they pursue various strategic initiatives, including partnerships, mergers & acquisitions, collaborations, and new product/ technology development. For instance, in January 2021, OTTO Motors announced a partnership with PULSE Integration, a material handling specialist, to deploy autonomous mobile robots for materials handling in manufacturing. AMRs were used to work in progress pallets, transport raw material pallets, and finished goods pallets in place of forklifts.

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COVID – 19 IMPACTS ON BUSINESS (IMPACT – HIGH)

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ADVANCEMENTS IN LATEST TECHNOLOGIES (IMPACT – HIGH)

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Global Key Company Ranking Analysis

Rank	Company name
1	KUKA AG
2	ABB
3	
4	
5	

Source: BJET, ETS, Education Technology Magazine, Design and Technology Education: An International Journal, IEEE, Company Annual Reports, Grand View Research

Material handling equipment are required in warehouse or manufacturing plant for quick movement of goods or products, locating stock, preparing orders and localizing stock. As industrial enterprises around the world adopt advanced manufacturing techniques, build new factories and warehouses, and upgrade their existing facilities, they purchase new automated storage, retrieval systems and other material handling equipment as they offer efficiency and productivity gains and reduce labor costs. Following is an analysis of the major AMR industry players:

KUKA AG designs and develops automation solutions for hospitals, warehouses, and distribution centers worldwide. The company operates through five business segments, namely Systems, Robotics, Swisslog, Swisslog Healthcare, and China. The robotics segment provides the industrial robots and core components for automating production processes.

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Company Geographical Presence

Company Name	Regional Presence				
	North America	Europe	Asia Pacific	South America	Middle East & Africa
KUKA AG					
Company 2					
Company 3					
Company 4					
Company 5					

Note: High (●) Medium (●) Low (●) No (●)

Section Description:

This section covers company profiles of top 10 vendors covering brief background about company, plant location & business type, financial performance, product benchmarking and strategic initiatives. Data related to M&A and divestures, product or service launches, awards & recognition and other operations/ expansions are listed.

Your interest in additional list of company profiles can be accommodated based on request.

Company Profiles

ABB

- ▶ **COMPANY OVERVIEW**
- ▶ **FINANCIAL PERFORMANCE**
- ▶ **PRODUCT BENCHMARKING**
- ▶ **RECENT DEVELOPMENTS**

Bleum

- ▶ **COMPANY OVERVIEW**
- ▶ **PRODUCT BENCHMARKING**
- ▶ **RECENT DEVELOPMENTS**

Boston Dynamics

- ▶ **COMPANY OVERVIEW**
- ▶ **PRODUCT BENCHMARKING**
- ▶ **RECENT DEVELOPMENTS**

Clearpath Robotics, Inc.

- ▶ **COMPANY OVERVIEW**
- ▶ **PRODUCT BENCHMARKING**
- ▶ **RECENT DEVELOPMENTS**

GreyOrange

- ▶ **COMPANY OVERVIEW**
- ▶ **PRODUCT BENCHMARKING**
- ▶ **RECENT DEVELOPMENTS**

Harvest Automation

- ▶ **COMPANY OVERVIEW**
- ▶ **PRODUCT BENCHMARKING**
- ▶ **RECENT DEVELOPMENTS**

IAM Robotics

- ▶ **COMPANY OVERVIEW**
- ▶ **PRODUCT BENCHMARKING**
- ▶ **RECENT DEVELOPMENTS**

inVia Robotics, Inc.

- ▶ **COMPANY OVERVIEW**
- ▶ **PRODUCT BENCHMARKING**
- ▶ **RECENT DEVELOPMENTS**

KUKA AG

- ▶ **COMPANY OVERVIEW**
- ▶ **FINANCIAL PERFORMANCE**
- ▶ **PRODUCT BENCHMARKING**
- ▶ **RECENT DEVELOPMENTS**

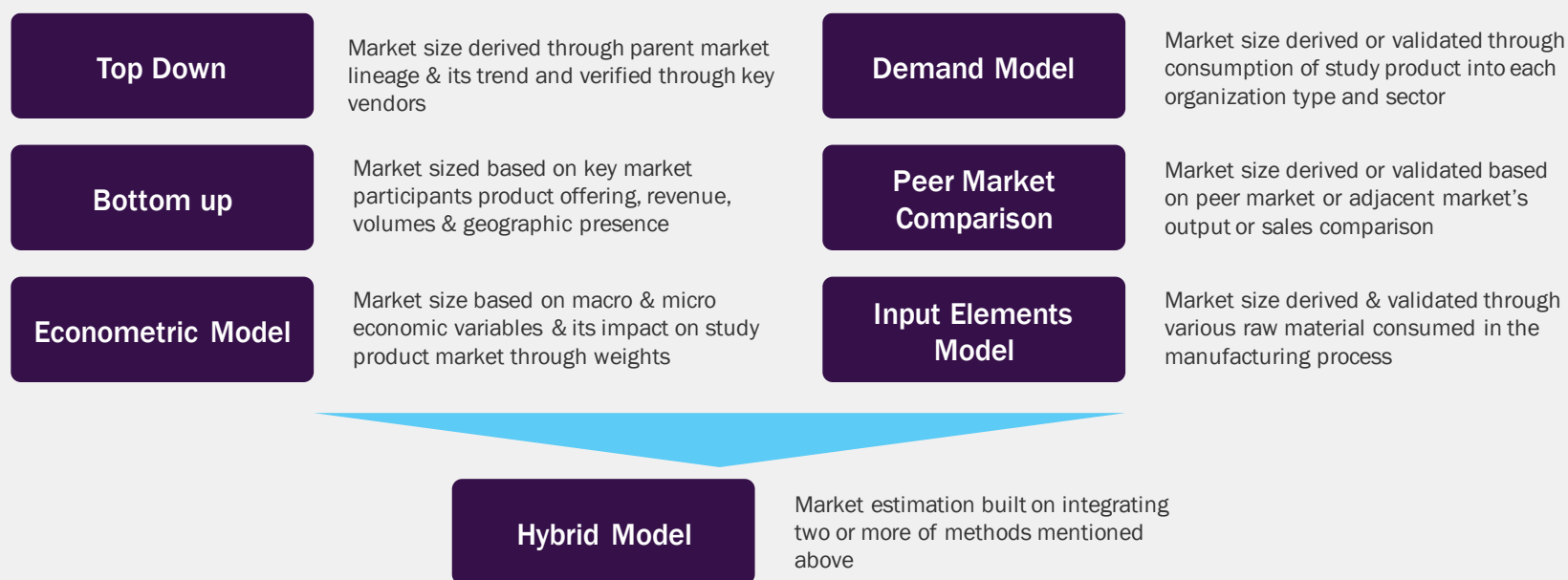
Teradyne Inc.

- ▶ **COMPANY OVERVIEW**
- ▶ **FINANCIAL PERFORMANCE**
- ▶ **PRODUCT BENCHMARKING**
- ▶ **RECENT DEVELOPMENTS**

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Methodology and Scope

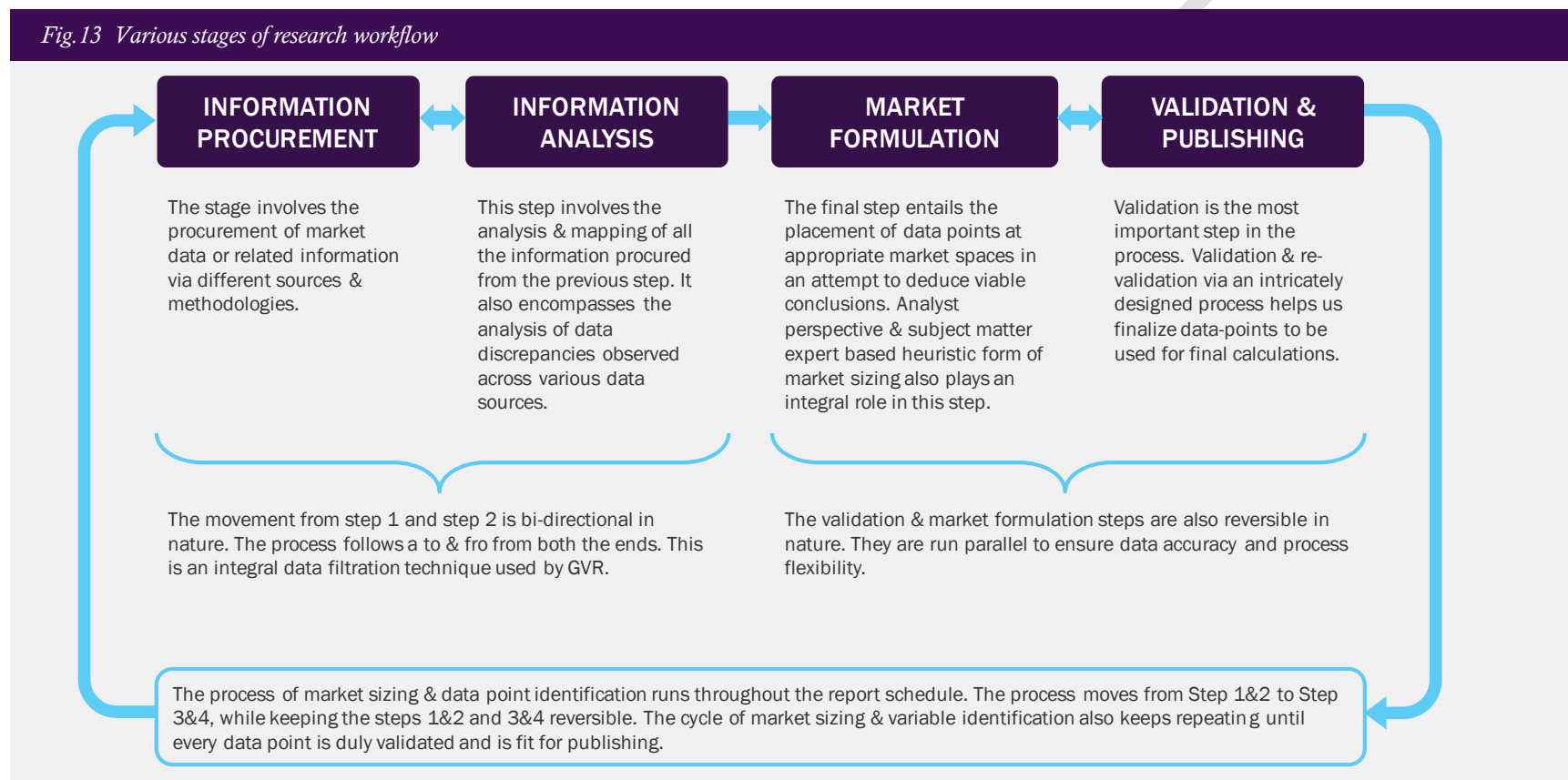
Fig.12 Methodology



Autonomous mobile robot (AMR) market report estimations are derived based on hybrid model leveraging top down and bottom-up approach tracking major market player's revenue and integrating information with peer market comparison model using other major autonomous mobile robot (AMR) category. The COVID-19 impact assessment for short term forecast, the financial performance analysis of the prominent autonomous mobile robot (AMR) industry players was used to determine the future outlook.

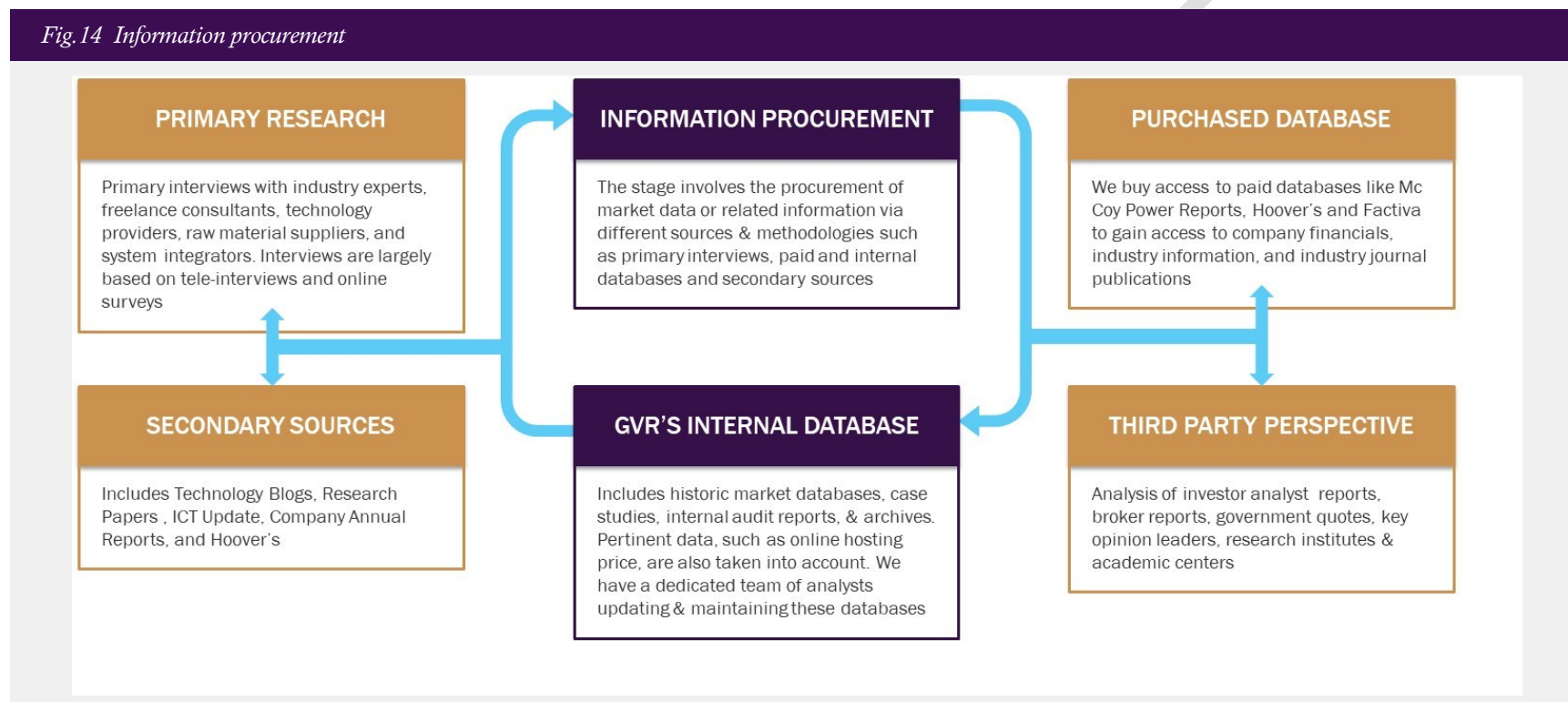
Research Workflow

Fig.13 Various stages of research workflow



Information Procurement

Fig.14 Information procurement



Information procurement is one of the most extensive stages in our research process. As illustrated in the figure above, the techniques can broadly be categorized into five sections, as stated below:

▶ **PURCHASED DATABASE**

- Includes company databases such as Factiva: This helps us compile metadata on historical sales, prices, company revenues, and other industry statistics. Also, it serves as an important step in market sizing, especially, in case of commodity-flow techniques
- Other sources include SME journals, pertinent databases from third-party vendors to gain insights into:
 - Usage rates
 - Potential market-related statistics
 - Information on unmet needs
 - Regional expenditure pattern
 - Investment information or opportunity-based statistics

▶ **GVR'S INTERNAL DATABASE**

- Includes our internal database of data points, collected as a result of previous research & studies and information made available via our database management team
- Also includes internal audit reports & archives

▶ **SECONDARY SOURCES**

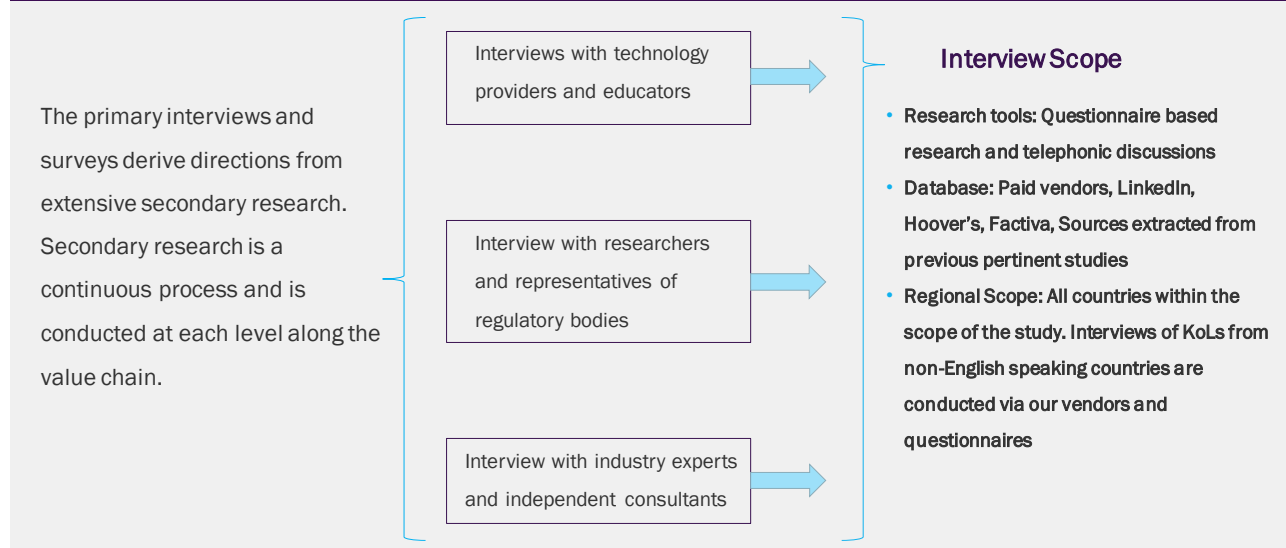
- A list of secondary sources along with the information extracted from them will be available in the final deliverable.
- Sources consulted during preliminary phase include Robotics Tomorrow, Automation Magazine, Manufacturing Tomorrow, Assembly Magazine, National Institute of Standard and Technology, MHI, Logistics Insight Asia, Technology Blogs, ICT Update, Company Annual Reports, and Hoover's, and Company Websites.
- Notable examples include white papers, government statistics published by organizations like Robotics Industries Association, US Legal, company filings, investor documents etc.
- Secondary databases are aimed at gathering market historical statistics, product classifications, and to understand trends available through public domains.

▶ **THIRD PARTY PERSPECTIVE**

- This section includes market derivation through investor analyst reports, broker reports, academic commentary, government quotes & wealth management publications.

PRIMARY RESEARCH

Fig.15 Primary research pattern



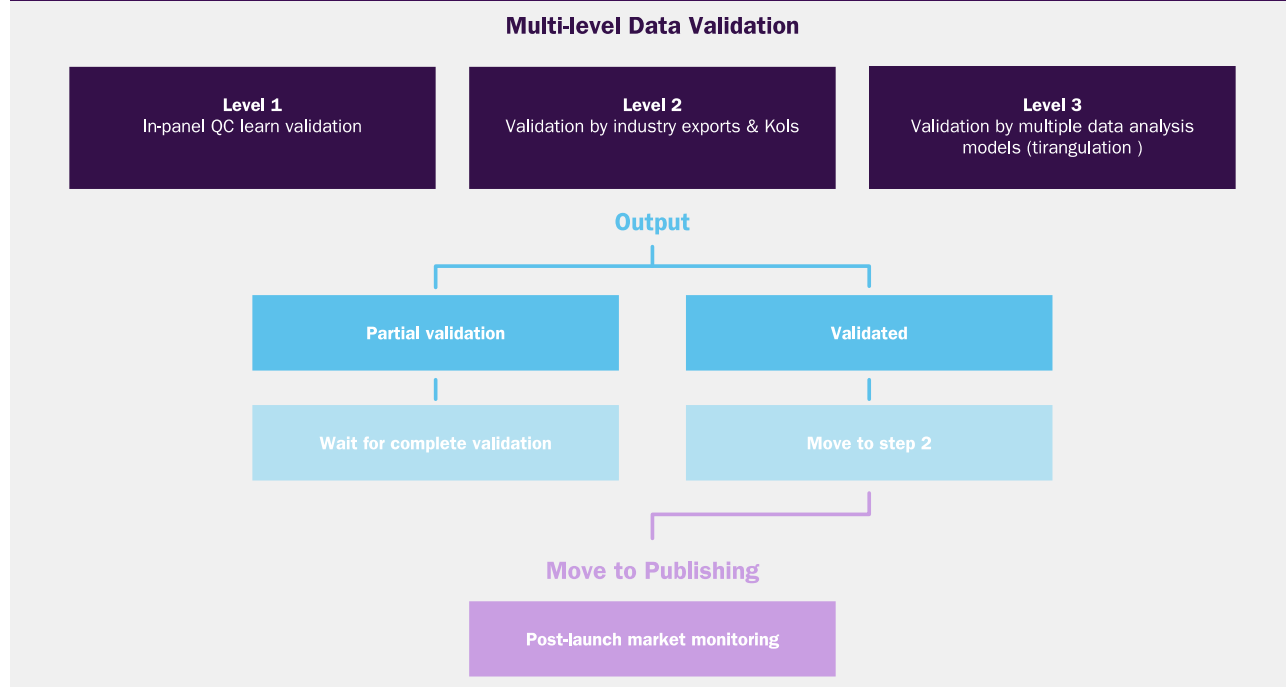
Information Analysis & Data Analysis Models

Information procured from secondary and primary initiatives are then, analyzed by using the following tools/ models: (a partial list)

- Identifying variables and establishing market impact
- Establishing market trends
- Analyzing future opportunities and market penetration rates by understanding product commercialization, regional expansion etc.
- Analyzing changes in the industry dynamics to establish future growth
- Analyzing sustainability strategies adhered by market participants to determine future course of the market
- Analyzing historical market trends and super-imposing them on the current and future variables to determine year-on-year trend
- Keeping a track of technological advancements in individual segments
- Base numbers are established by analyzing the following:
 - Company revenues and market share (this list generally includes the analysis of revenue published by publicly listed manufacturers)
 - Derivation of market estimates via analyzing parent and ancillary markets
- Model selection: demand-based bottom-up approach and mixed approach (top down and bottom up)

▶ **MARKET FORMULATION & VALIDATION**

Fig.16 Primary research pattern



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- ▶ Clients have free access to pre-sale analyst briefs to discuss requirements and recommendations

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- ▶ Yes, we sell sections of our reports
- ▶ You also have the option to buy excel & PPT versions of our reports.
- ▶ You may buy customized market intelligence based on your budget. We have a product to offer irrespective of the price point

How do you arrive at these market numbers?

- ▶ Our research methodology is a three-step cyclic process:
- ▶ It starts with Information Procurement from internal DBs, paid primary and secondary sources
- ▶ The second step is Data Analysis and Modeling
- ▶ The third step is by Data Validation via industry expert opinions
- ▶ For a detailed research methodology, please request for a sample report.

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Related Reports

Our Expertise:

This Autonomous Mobile Robot (AMR) report is part of a bundle of reports on next-generation technologies. We have published a total of 30+ reports in this space, and plan to continue publishing data books and reports on this subject. You can access all these data books & reports by subscribing to our pay-as-you-go model.

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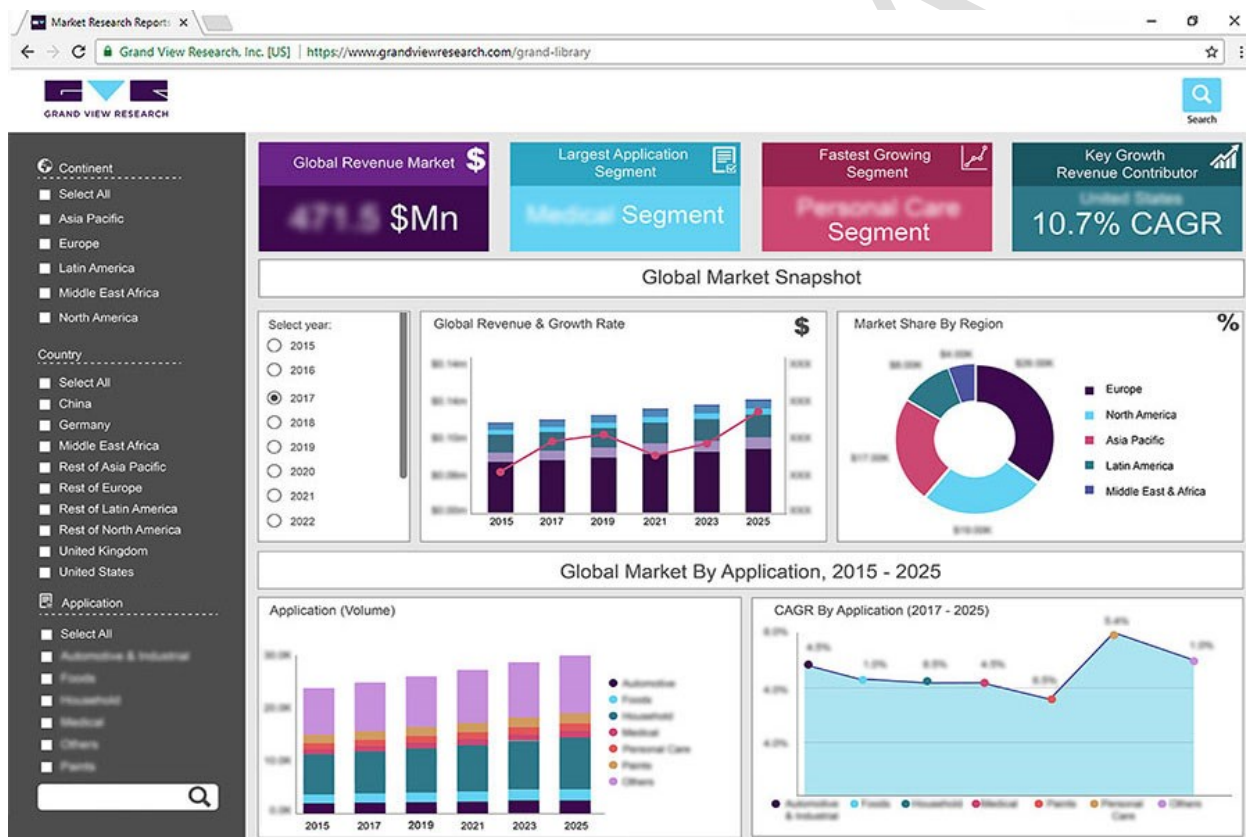
- [Automated Guided Vehicle Market Size, Share & Trends Analysis Report By Vehicle Type, By Navigation Technology, By Application, By End Use Industry, By Component, By Battery Type, And Segment Forecasts, 2022 – 2030](#)
- [Warehouse Robotics Market Size, Share & Trends Analysis Report By Product \(Mobile Robots, Cartesian Robots\), By Function, By Payload Capacity, By Software, By Application, And Segment Forecasts, 2019 - 2025](#)

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Sample Figure:



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